Integrative Multi-Omics Machine Learning Models for Breast Cancer Stage Prediction Using TCGA Data

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Abstract

Breast cancer is a complex disease and accurate stage prediction is crucial for prognosis and early treatment. This project explores the use of RNA-seq and DNA methylation data from The Cancer Genome Atlas (TCGA) to build machine learning models for predicting cancer stage. The primary goal was to assess whether models trained on gene expression and methylation data can accurately predict cancer stage, and how their performance compares to models using only clinical phenotype data and combined clinical and multi-omics data. Additionally, we aimed to identify biologically relevant features associated with disease progression. The dataset was further preprocessed through filtration, imputation and z-score transformation. After this we splitted the data into 80:20 train-test ratio. Differential methylation and expression analyses were conducted on the training data using Kruskal-Wallis and Dunn's post-hoc tests. Machine learning models, including Random Forest, Support Vector Machines (SVM) and Logistic Regression, were trained using the differential analysis results. Finally, gene set enrichment analysis (GSEA) and over-representation analysis (ORA) were applied to annotate the most important features. Our results indicate that models trained solely on clinical phenotype data performed best, followed by models incorporating both clinical and multi-omics data. Models trained without clinical data performed the worst. Enrichment analysis revealed pathways potentially linked to disease progression, highlighting the added biological insight from interpreting selected features. These findings suggest that, at least in our current setup, multi-omics data without clinical context may not carry sufficient signal for accurate breast cancer stage classification.

Keywords: Machine Learning, Breast Cancer, Diagnosis, Cancer Stage Prediction, Multi-Omics, RNA-seq data, Methylation data, Integrative analysis

1 Background

Breast cancer is a highly prevalent and complex disease and remains one of the leading causes of cancer-related deaths among women worldwide [1]. Determining treatment plans and evaluating prognosis depend on accurate breast cancer stage prediction at diagnosis [2]. Conventional stage prediction is highly dependent on clinical variables such as tumor size, lymph node involvement, and metastasis status.

Multi-omics approaches, which integrate diverse molecular layers offer the potential to unravel complex biological mechanisms underlying cancer progression [3]. The Cancer Genome Atlas (TCGA) provides a multi-omics resource for several cancer types, including breast cancer, enabling the development of integrative computational models for prediction and biomarker discovery. However, integrating heterogeneous omics data poses methodological challenges due to differences in scale, distribution, and noise, as well as high dimensionality and sample size limitations [4].

In the context of cancer classification and prognosis, several studies have evaluated multi-omics machine learning pipelines. For example, Poirion et al. developed a semi-supervised framework (DeepProg) that integrates RNA-seq, methylation, and miRNA data to predict survival across multiple cancer types [5]. Other work by Mohamed and Ezugwu used a deep learning model to integrate omics data for lung cancer classification, demonstrating high accuracy when multiple data types were fused [6]. Furthermore, Yang et al. applied machine learning models to predict colorectal cancer survival using data from the microbiome, miRNA and tissue microbiome, finding that the microbiome data alone achieved the highest predictive power [7]. While these studies illustrate the potential of multi-omics integration, their outcomes also highlight that predictive performance varies considerably depending on the cancer type, data quality, and the availability of clinical variables.

In the context of breast cancer, omics-based modeling has particular promise due to the well-characterized molecular subtypes (e.g., basal-like, luminal A/B, HER2-enriched) and their differing prognoses [8, 9]. Gene expression profiles have been used successfully for subtype classification, risk stratification, and treatment selection (e.g., Oncotype DX, MammaPrint) [9].

In this project, we explored whether gene expression and DNA methylation data from TCGA breast cancer samples can be used to predict cancer stage using supervised machine learning. Our aim was to assess the contribution of omics data alone and in combination with clinical phenotype data to classification performance. We also sought to determine whether any of the top-ranked features from our models, selected based on their contribution to classification, showed known associations with breast cancer biology.

2 Goal

The primary goal of this project was to evaluate whether gene expression and DNA methylation data alone can be used to accurately predict breast cancer stage using machine learning models. Since cancer stage is typically determined using clinical and pathological criteria, we aimed to test the predictive value of molecular data independently of those features (Figure 1). By doing so, we assessed whether omics data carry intrinsic biological signals that correspond to cancer progression. In addition, we compared the performance of models trained with omics data alone to those incorporating clinical phenotype data, and to models trained solely on clinical data, to determine the extent to which clinical information improves classification.

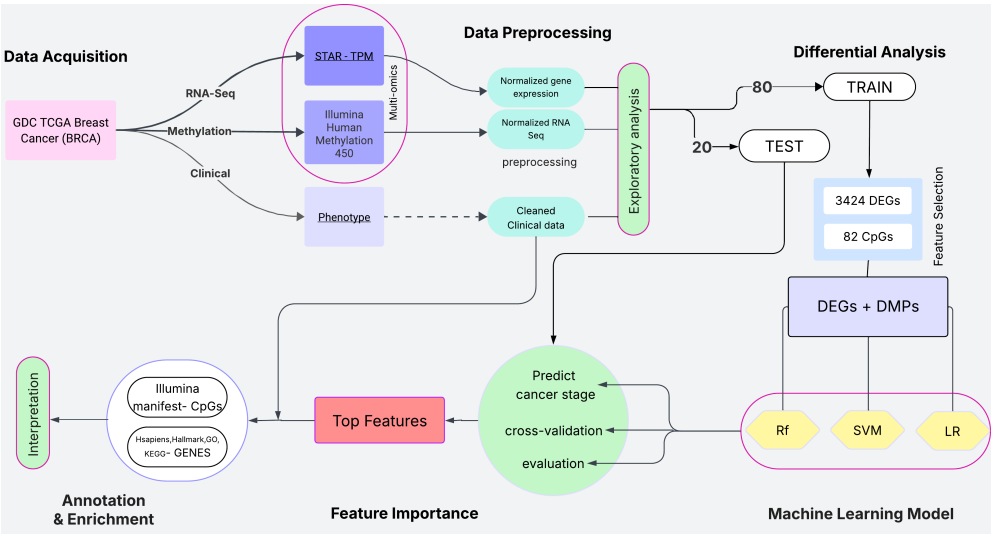


Fig. 1 Overview of the analytical pipeline used in this study. The workflow consists of preprocessing of omics and clincial data, followed by exploratory analysis, differential analysis, data partitioning, feature selection, and supervised machine learning model development.

3 Data and Preprocessing

3.1 Data Acquisition

For this project, we acquired the multi omics data from the TCGA BRCA Project and accessed it directly from the Xena browser [10]. Initial composition contains around 24 datasets. The shortlisted datasets for our design were gene expression data(RNA-seq, STAR-TPM), DNA methylation data (Illumina Human Methylation 450) and clinical data(Phenotype) from the 1200 patient samples diagnosed with breast cancer, giving a comprehensive understructure for an integrative machine learning analysis as shown in Table 1. The common amples across these data were 770 and ensured multi-modality.

Data	Samples	Type	Initial size	Preprocessed size
RNA-Seq	1226	STAR-TPM	60000	20000
Methylation	893	Illumina Human Methylation 450	500000	190000
Clinical	1255	phenotype	80 columns	9 columns

Table 1 Summary of multi-omics data types and corresponding sample counts considered for the study.

The Gene expression data from the TCGA BRCA cohort has been derived from RNA sequencing method of primary tumor samples, these are quantified and normalized using TPM with subsequent log transformations. The gene expression dataset consists of 1226 samples. This is a high-throughput transcriptome profiling that includes thousands of genes per sample and identifies direct functional activity inside malignancies [11].

The methylation dataset is obtained using the Illumina Infinium HumanMethylation450 BeadChip, profiling over 450,000 CpG sites genome-wide via quantitative beta values ranging from 0 (unmethylated) to 1 (completely methylated)(ref). There were total of 893 samples in the DNA methylation dataset. This encompasses a comprehensive and epigenomic range with an expected bimodal beta- value distribution of CpG sites, and is also an important component of TCGA epigenetic studies in various research papers [12]

Lastly the clinical phenotype dataset contains patient metadata such as tumor stage, hormone and receptor status, patient age, diagnostics, tumor demographics and survival-related features. The Cancer Imaging Archive (TCIA) uses TCGA's standardised clinical forms for all cancer types to ensure consistent annotations. There were a total of 1255 samples in the phenotype dataset. This information is crucial for predicting cancer stage and determining how omics data improves model performance beyond traditional clinical variables [13]

3.2 Preprocessing

Each category of data was partially preprocessed and then further preprocessed independently. For methylation data, only the autosomal chromosomes were included and probes containing more than 20% missing values were removed [14]. Probes that mapped ambiguously to numerous genes, as well as those with very low signal intensity (determined by a minimum counts-per-million criteria), were excluded. Additionally the missing values were estimated using the median beta value for each CpG site. As expected, beta values were distributed bimodally as shown in Figure 2.

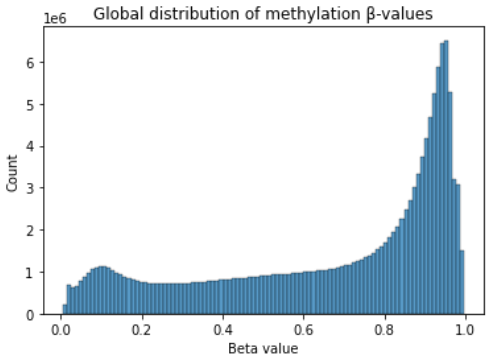


Fig. 2 A visual representation of preprocessed methylation data with beta values expression

To prevent any sex-related confounding effects, the gene expression data was obtained only from female subjects. Genes found on the X and Y chromosomes were eliminated. Genes with low expression levels (TPM less than 1 in more than 20% of samples) and zero standard deviations (i.e., constant expression across all samples) were excluded [15]. The expression matrix was then normalized using Z-score normalisation across genes, which is visible in Figure 3. The remaining missing values were validated for consistency.

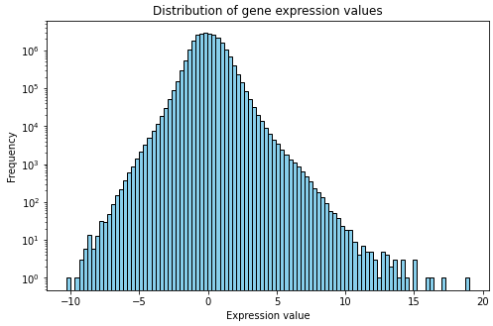


Fig. 3 A visual representation of preprocessed gene expression data showcasing normal distribution

Lastly, since the clinical data contained too many features as columns, we ensured to stick closely to the features related to stage prediction. Only columns relevant to disease staging and tumor characteristics (e.g., tumor size, stage, tissue type) were retained. Samples with missing or unknown tumor stage annotations were from the dataset to deal with null value issues . Sub-stages (e.g., IIA, IIB) were merged into more general stages (e.g., Stage II) to facilitate the categorization, normal (or non-tumor) samples and multiple entries were not included as shown in Figure 4. The final dataset was examined for class imbalance, particularly across tumor stages, which was addressed during model development [15].

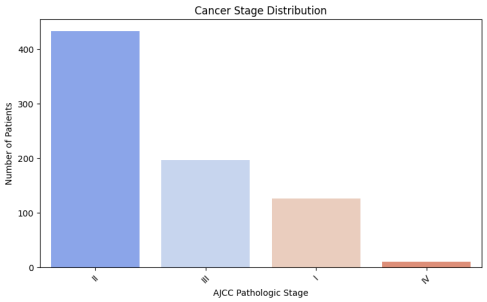


Fig. 4 Preprocessed clinical data showing sample distribution in each stage

4 Methods

4.1 Quality control and Exploratory Data Analysis

Once the data have been fully preprocessed, the next major step was to ensure our data is fit for downstream analysis, which was done in the R environment. Quality control and data quality assessment were performed to check the quality of the data using the *dplyr, ggplot2, pheatmap, RColorBrewer* libraries. This process helps detect and correct technical artifacts, errors, or any biases before the analysis, and the focus is over data quality and reliability. Along with these, we also utilised dimensionality reduction techniques which included Principal Component Analysis (PCA), t-distributed Stochastic Neighbor Embedding (t-SNE), and Uniform Manifold Approximation and Projection (UMAP), which served as a factor for both quality assurance and data exploration. In our project, they were an essential step to verify structure and integrity of data.

Firstly, we did a quality control check for gene expression data using a general heatmap, which is a visual representation of our data where values are depicted by different shades shown in Figure [16]. This heatmap helped us visualize the any similarity or dissimilarity between samples based on gene expression of samples, however very few of them were closely related and the overall projection did not show any abnormal trend Figure 5. This was done by clustering rows and columns based on correlation of the filtered samples using Pearson correlation to expression similarities in the samples.

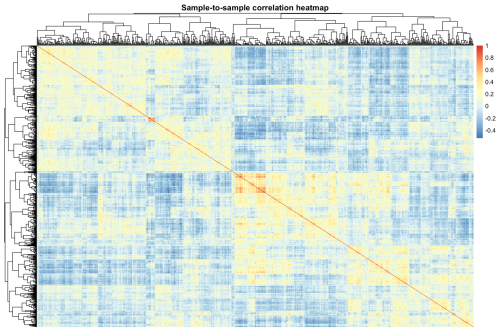


Fig. 5 A heatmap depicting the sample to sample correlation based on clustering and average gene expression values

Furthermore, a scree plot was plotted to define the proportion of variances explained by principal components followed by a PCA plot shown in Figure 6. PCA is a powerful dimensionality reduction technique used to transform traditional features into a new set of uncorrelated variables, and in our analysis we used sample clusters and checked for any potential outliers and batch effects [17]. In our analysis, scree plot helped us determine whether the initial PCs captured the most variance, hence suggesting structured and interpretable data, ensuring our data was free of noise and misleading information Figure 7.

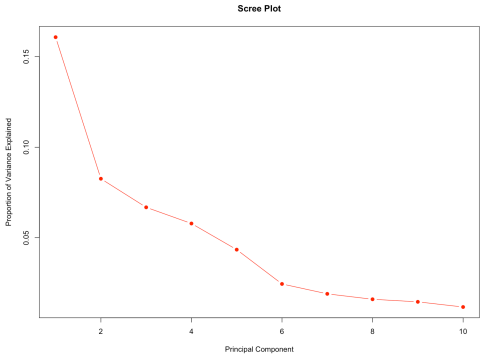


Fig. 6 Scree plot defining the proportion of variances explained by principal components

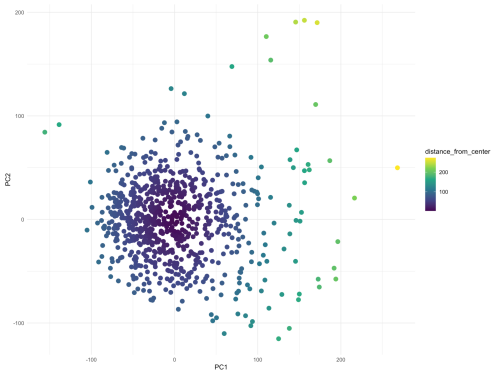


Fig. 7 PCA plot using the first two Principal Components explaining the proportion of variances in the gene expression data

In the Methylation data, violin plots were utilised to examine the beta value distribution stratified by the cancer stages, in our case we had four cancer stages initially, which checked for the stage-wise consistency Figure 8. Further, a bimodal distribution indicates preserved biological methylation signal as seen before in Figure 2 . This confirmed the expected pattern bimodality of the data, reflecting hypo-methylated and hyper-methylated regions across the genome. [18].

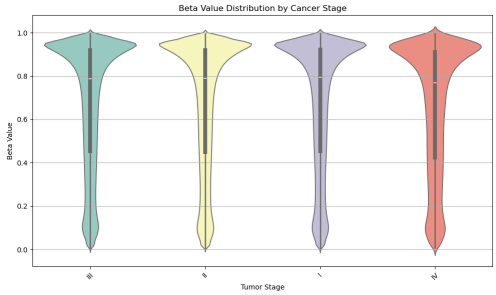


Fig. 8 A violin plot showcasing stage-wise consistency of methylation data

After the clinical data was filtered and cleaned according to our project idea, a significant class imbalance was observed. Stage IV consisted of very few sample ids. This was kept in notice and further resolved using the SMOTE tool. The Synthetic Minority Oversampling Technique (SMOTE) is a statistical technique that enhances the number of cases in your dataset in an equitable way. SMOTE creates synthetic exa,ples of underrepresented classes and balances class distribution without duplication. [19]. In our data, SMOTE was highly useful to address class imbalance in tumor stage labels, since we had few samples in stage III and IV and further created synthetic samples for the minor classes Figure 9. This method was applied only on training set of data and it improved predictive performance of the models.

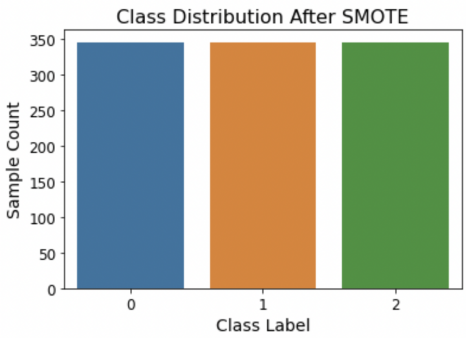


Fig. 9 Oversampling resolved using SMOTE tool further sorting class imbalance issues

Finally in order to check overall structure of data, identify outliers, or batch effects and comprehend the global distribution of expressed values, an exploratory analysis of each omics data type was performed [20]. The t-SNE or t-distributed Stochastic Neighbor Embedding is a non-linear, dimensionality reduction method which transforms high dimensional data into a lower dimensional space. It is significantly effective for preserving the local structure of the data and revealing natural clusters based on any similarities. Uniform Manifold Approximation and Projection (UMAP) on the other hand is another method that is faster and has better preservation of the global structure of our data. UMAP builds a graph that captures local relationships in the high-dimensional data, then optimizes a low-dimensional embedding so that the graph structure is preserved as closely as possible [21].

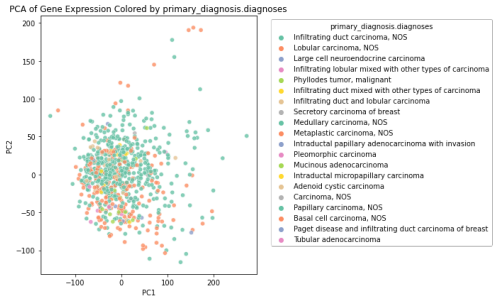


Fig. 10 Principal Component Analysis of Methylation profiles by the clinical primary diagnosis of the data

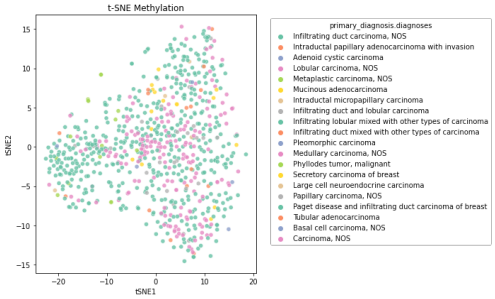


Fig. 11 Exploratory data analysis of methylation profiles with respect to primary diagnosis using dimensionality reduction technique t-SNE revealing distinct structure of data

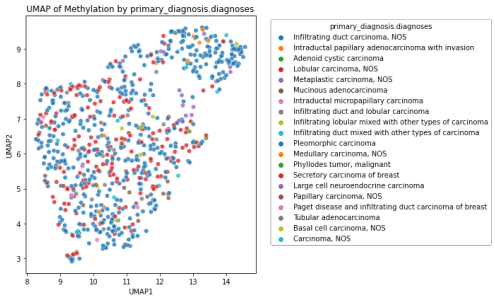


Fig. 12 Exploratory data analysis of methylation profiles with respect to primary diagnosis using dimensionality reduction technique UMAP revealing global structure of data

For Methylation data, these dimensionality reduction methods (t-SNE, PCA and UMAP) were used with clinical annotations (mainly primary diagnosis) apart from the stage labels to check for potential batch effects or clusters of samples by cancer subtype and outliers Figure 10 , 11 and 12. No particular trends were observed in the plots as the samples were spread evenly across the graph with no significant shape or cluster.

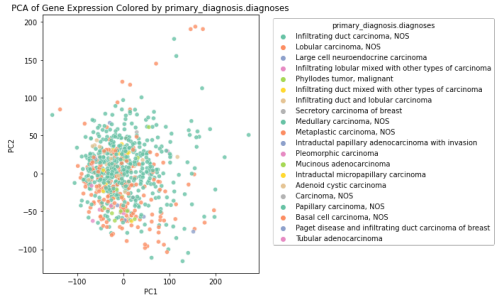


Fig. 13 A PCA with reduced dimensionality showcasing variance structure across samples. Points are colored by condition of primary diagnosis or subtypes for any reveal global trends.

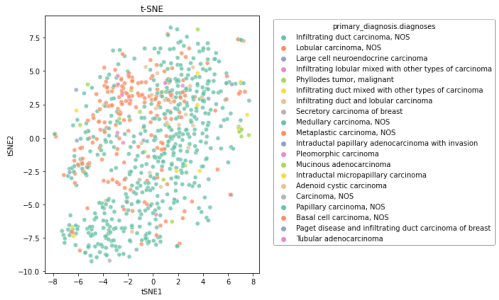


Fig. 14 t-SNE done on gene expression data reveals potential sample clusters by highlighting local trends in the data.

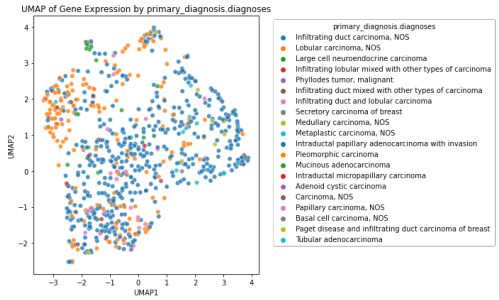


Fig. 15 A UMAP plot projecting potential cluster identification by showcasing global trends in the structure of gene expression data.

For the gene expression data, the first plot was a principal component analysis (PCA) plot which was performed over the z-score normalized expression matrix in Figure 13, followed by the t-SNE in Figure 14 and UMAP plot which is Figure 15 to visualize potential batch effects, clustering of profiles based on clinical subtypes.

With the help of these methods, we analyzed data completeness, variability and any potential confounding factors, which acted as a guiding step for the differential analysis and feature selection. Similar to methylation data, gene expression data also did not possess any significant trends or clusters and hence no batch effects were indicated in this analysis.

4.2 Data Partitioning

To ensure a robust evaluation of all three models and prevent any data leakage, the complete dataset was split into training and testing sets prior to differential analysis, feature selection or model development. A typical 80:20 split was done, with 80% of the data used for training and 20% set aside as an independent test set for later. In order to reduce potential bias, we randomly splitted the data to ensure that both subsets reflect the underlying distribution of cancer stages [22].

This technique permitted the formation of models based solely on training data, while also giving an unbiased evaluation of generalization performance on unseen test samples [22]. Furthermore, executing differential expression and methylation analyses only on the training set reduced data leakage, protecting the integrity of feature selection and downstream machine learning outcomes [23].

4.3 Differential Analysis

Differential analysis is a statistical technique to identify features whose expression or methylation levels can change between different biological circumstances or phenotypes, for example (Stage I vs Stage II of breast cancer). It is considered to be one of the most important tool in high-throughput data processing as it screens the genetic changes or expression that can be linked to disease progression, subtype differences, or therapy response. For seamless class comparison, stage III and IV were combined.

4.3.1 Differential Gene Expression analysis

In order to detect the significant genes in cancer stage progression, we firstly used a non-parametric Kruskal-Wallis test. This is a type of test that ranks gene expression levels across all samples and does a comparison of average ranks among the groups (stages) to identify statistically significant differences in expression [24]. Since it does not require a normal distribution, it is an appropriate choice for biological data even if it has outliers or bias.

Dunn's Post-hoc test was performed to compare individual tumor stages. This particular test is accountable for multiple testing scenarios and helps determine if the stage pairs (e.g., Stage I with Stage II) demonstrate any significant differences in gene expression as shown in Figure 16. With this type of approach, we successfully found 3,424 genes that were differentially expressed throughout tumor stages with a false discovery rate (FDR) of less than 0.05, which compensates for type I errors caused through multiple comparisons [25].

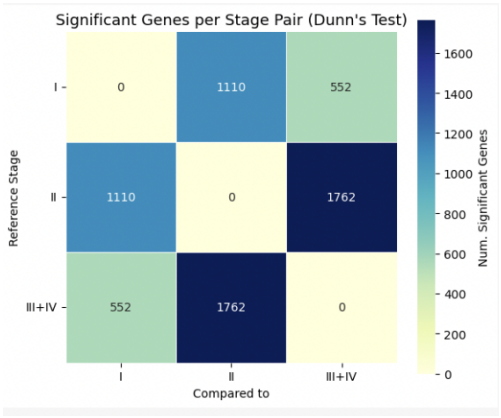


Fig. 16 A confusion matrix with significant genes per Stage pair curated by Dunn's test

4.3.2 Differential Methylation analysis

In preprocessed methylation data, we began by converting the beta values (with proportions between 0 and 1 that will signify the methylation levels) into M values using a logit transformation. This adjustment was done since beta values have heterogenetic variances (i.e., non-constant extreme values near 0 or 1 also known as heteroskedasticity) [26]. It was derived that such characteristics violate assumptions made by statistical models, in our case linear model with normality and homoscedastic data. Hence, converting these beta values to M-values makes data unrestricted and symmetrically distributed. This further enhances statistical reliability of the differential methylation analysis and makes it suitable for our linear model [26].

After this, we labeled tumor stages as a categorical variable. Along with accounting variances for both within and between the groups, linear modeling assesses correlation between levels of methylation and the cancer stages. We used the empirical Bayes moderation from limma framework to improve variance estimation. This is done by shrinking variances of probe wise sample towards a common value. This results in reduced risk of variance overstimulation due to any noise [27].

Finally if the CpG sites had a FDR less than 0.1, they were determined to be important and helpful for machine learning models to make correct predictions. The p-value distribution leans towards left (Figure 17), indicating that there are potential important sites. The FDR threshold was set to 0.1 and not 0.05 as no significant sites were captured in that threshold as shown in the volcano plot in Figure 18. At the end, 82 CpG sites were identified to be potentially important across different tumor stages. The Q–Q plot in Figure 19 was used to check the normality of M-values, as assumed by the linear model. We did not observe any significantly non-normal trend.

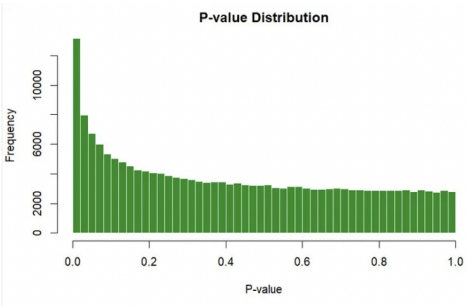


Fig. 17 P-value distribution table of differentially methylated regions

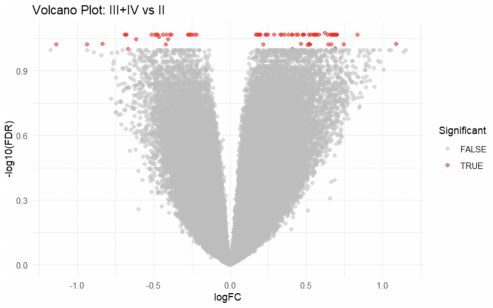


Fig. 18 Volcano plot displaying differential methylation results across tumor stages. Each point represents a CpG site, plotted based on the log2 fold change in methylation (x-axis) and the adjusted p-value (y-axis). The red points denote differentially methylated CpGs (FDR ¡ 0.1), highlighting loci with the most important changes in methylation between cancer stages.

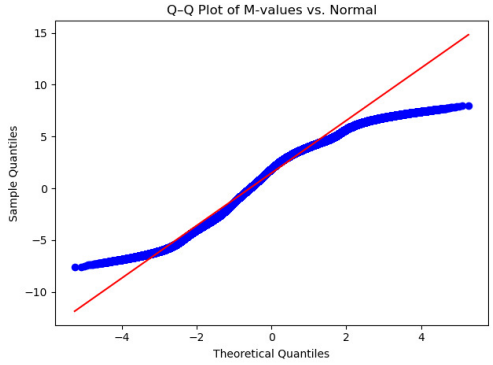


Fig. 19 Q-Q Plot to check the normality of M-values

4.4 Scaling and Feature Selection

In order to make sure that the features were on a comparable level with the each model sensitivity with respect to the input values, a scaling standardization was applied, known as StandardScaler [28]. By doing this, we ensured that our data is transformed in a way that each feature has a mean of zero and a standard deviation of one. This process helps stabilizing and optimizing the performances of all three models [29]. This standardization was particularly applied on the training subset and thereafter consistently used to transform the rest of the test subset, ensuring to avoid any data leakage [30].

In order to minimize the dimensionality and preserve the most informative variables, we then carried out feature selection, an important aspect of machine learning model. We employed the **SelectKBest** approach, which is a feature selection method using statistical tests to analyze each feature separately and determine how it relates to the defined target variable [31]. We sought to balance model complexity and performance by experimenting with various values of k. By lowering the chance of overfitting, this procedure increased generalizability, reduced noise, and increased computing efficiency [30].

4.5 Machine Learning Models

We utilized three different machine learning models and compared their performances. We chose methods capable of multi-class classification. We performed 5-fold cross-validation and selected the best hyperparameters for each model based on validation performance.

4.5.1 Logistic Regression

Logistic regression models the probability that a sample belongs to a particular class, in our case, a cancer stage. It is capable of multi-class classification. It's highly interpretable, since inspecting the coefficients reveals how each feature contributes to the model. However, logistic regression assumes linear decision boundaries and may struggle with complex patterns of omics data. [32]

4.5.2 Random Forest

Random forest is an ensemble learning method that builds a collection of decision trees and aggregates their predictions to make a final decision. It naturally supports multi-class classification, as each tree votes for a class and the majority vote determines the final prediction. While it is less interpretable compared to logistic regression, random forest can capture non-linear relationships in the data. Feature importance metrics can still provide insight into which variables contribute most to the model's predictions. [33]

4.5.3 Support Vector Machine (Linear kernel)

Support vector machine (SVM) with a linear kernel is a supervised learning model that aims to find an optimal hyperplane that best separates the classes by maximizing the

margin between them. By default, a one-vs-rest strategy is used to tackle the multi-class problem, where a separate binary classifier is trained for each class against all others. It assumes linear separability and may not capture non-linear patterns in the data. [34]

4.6 Annotation of Selected features

To gain biological insight into the features prioritized by our machine learning models, we performed annotation and pathway enrichment analysis. Interpreting the biological relevance of selected features can help contextualize model outputs and potentially uncover mechanisms related to cancer progression. For this analysis, we used the top 30 most important features from each of the models trained exclusively on omics data. Feature importance was determined based on model-specific criteria: absolute coefficients for Logistic Regression, mean decrease in impurity for Random Forest and quantification of feature discrimination by comparison of inter- and intra-class variance for Support Vector Machine (SVM). We first separated the features into CpG sites and genes for each model. CpG sites were mapped to their associated genes using the *minfi* R package, based on Illumina 450k annotation data. For gene expression features, Ensembl IDs were matched to HGNC gene symbols using the *biomaRt* package to improve interpretability.

4.6.1 Pathway enrichment

To investigate the functional relevance of the selected gene features, we performed pathway enrichment analysis using both Overrepresentation Analysis (ORA) and Gene Set Enrichment Analysis (GSEA). ORA was conducted using three gene set databases: Hallmark gene sets from MSigDB, Gene Ontology Biological Processes (GO BP) and KEGG pathways. It works by testing whether a predefined set of genes is overrepresented among a list of selected features (e.g., top-ranking genes), compared to what would be expected by chance, using statistical tests such as the hypergeometric test [35]. For GSEA, we used only the Hallmark gene sets. GSEA detects coordinated changes in expression or significance without the need for arbitrary cutoffs by determining whether the members of a gene set tend to be towards the top or bottom of a sorted list of all genes from the dataset [36]. The enrichment analyses were performed separately for the top 30 gene features from each machine learning model trained on omics data.

In the ORA results, pathways with statistically significant adjusted p-values were marked. For GSEA, where no pathway reached statistical significance, we highlighted the pathway with the highest enrichment score. Additionally, we identified and flagged leading genes that appeared in multiple enriched pathways across models. These pathways and genes were then further investigated through literature research to explore their potential relevance to breast cancer biology and progression.

5 Results

In this section, we present the performance results of the machine learning models we trained. Additionally, we report the outcomes of the annotation analysis performed on

the top 30 features extracted from each of the three models, Support Vector Machine (SVM), Logistic Regression and Random Forest, trained using omics data only (gene expression and DNA Methylation).

5.1 Machine learning model performance comparisons

5.1.1 Models trained with omics data only

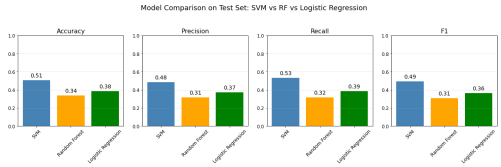


Fig. 20 Performance metrics of machine learning models trained using omics data only.

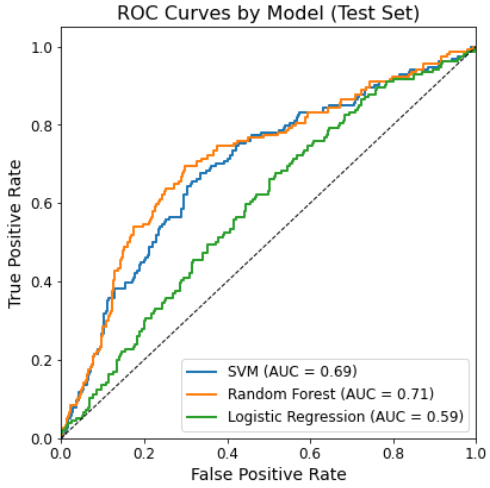


Fig. 21 ROC curves of machine learning models trained using omics data only.

The performance metrics of the machine learning models trained exclusively on omics data (gene expression and DNA methylation) are shown in Figure 20. All three classifiers—Support Vector Machine (SVM), Logistic Regression and Random Forest—achieved low performance across accuracy, precision, recall and F1-score. SVM showed slightly higher values overall, with accuracy reaching 0.53 and F1-score at 0.38, while Random Forest and Logistic Regression consistently performed below 0.4 across most metrics. These results suggest limited predictive power when using omics data alone.

The corresponding ROC curves for the same models are shown in Figure 21. Random Forest achieved the highest AUC at 0.71, followed by SVM at 0.69 and Logistic

Regression at 0.59. Although slightly better than random chance, the ROC curves confirm the overall weak discriminative ability of the models when clinical data is excluded.

These figures illustrate that gene expression and methylation features, in isolation, do not carry strong stage-predictive signals in this setup.

5.1.2 Models trained with omics and clinical data

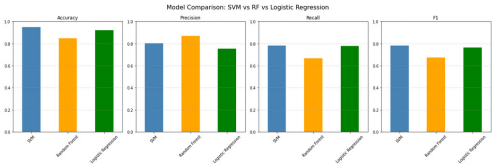


Fig. 22 Performance metrics of machine learning models trained using clinical and omics data.

Figure 22 presents the performance metrics of the machine learning models trained using both omics data and clinical features. Support Vector Machine (SVM) achieved the highest accuracy, reaching close to 0.95, followed by Logistic Regression and Random Forest. In terms of precision, Random Forest slightly outperformed the other models, while SVM and Logistic Regression showed comparable values. Recall was highest for SVM and Logistic Regression, with Random Forest performing slightly lower. F1-scores followed a similar trend, with SVM and Logistic Regression showing nearly identical results, both higher than Random Forest. Overall, all three models performed well when combining omics and clinical data, with SVM showing the strongest overall balance across all four metrics.

5.1.3 Models trained with clinical data only

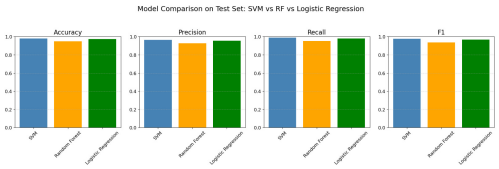


Fig. 23 Performance metrics of machine learning models trained using clinical data only.

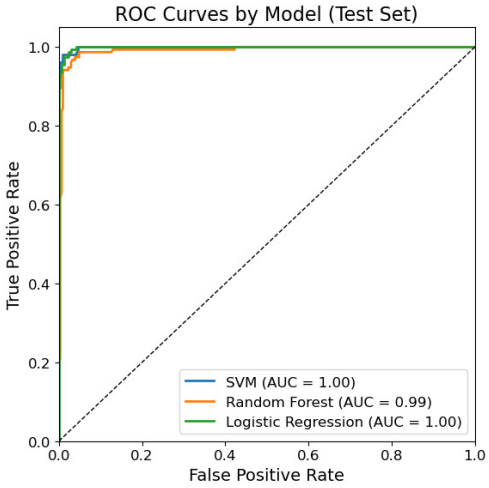


Fig. 24 ROC curves of machine learning models trained using clinical data only.

All three machine learning models trained solely on clinical data achieved the highest performance compared to models using omics-only or combined data. As shown in Figure 23, SVM achieved the highest accuracy, precision, and recall, followed closely by Logistic Regression and Random Forest came third with metrics all above 0.85. The corresponding ROC curves 24 show consistent class separability across models, with Logistic Regression and SVM displaying AUC of 1.00, indicating perfect class separability. Overall, the performance metrics and ROC curves indicate stable classification results across all three models when trained exclusively on clinical variables.

5.2 Feature annotation and enrichment analysis

We selected the top 30 features of all three machine learning models that were trained with just the omics data. There were a mix of CpG-sites and genes in the feature lists, with usually more genes than CpG-sites present. We mapped the CpG-sites in the lists to their corresponding genes and utilized the genes to find enriched pathways. From the enriched pathways, we also highlighted genes of potential importance based on their roles in enriched pathways.

5.2.1 ORA

We found one significantly enriched pathway using the GO biological processes and Hallmark gene sets, which were the **bronchus development** pathway and the **interferon alpha response** pathway, respectively.

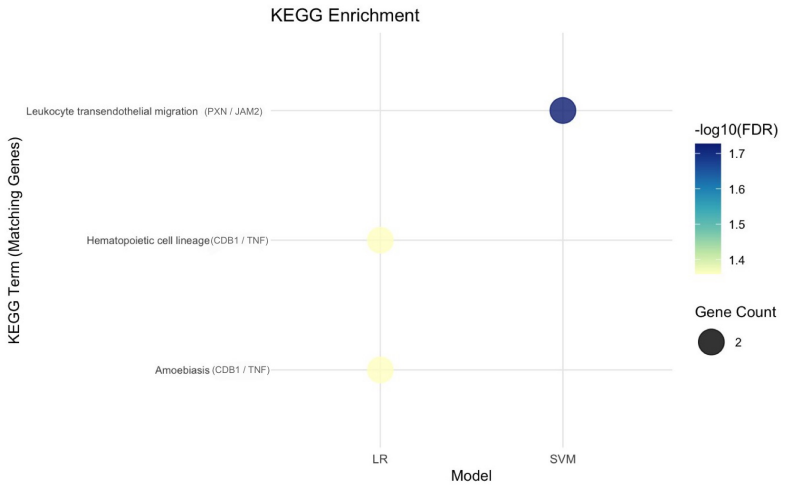


Fig. 25 Significantly enriched pathways found using KEGG with ORA.

Three significantly enriched pathways were identified using the KEGG Pathway resource (Figure 25). Among them, **Leukocyte transendothelial migration**, enriched from the feature list of the SVM model, showed the most significant false discovery rate (FDR) value. The other two pathways, **Hematopoietic cell lineage** and **Amoebiasis**, were identified based on features selected by Logistic Regression.

5.2.2 GSEA

GSEA was done using Hallmark gene set, to determine if it could identify pathways that were not identified by the ORA.

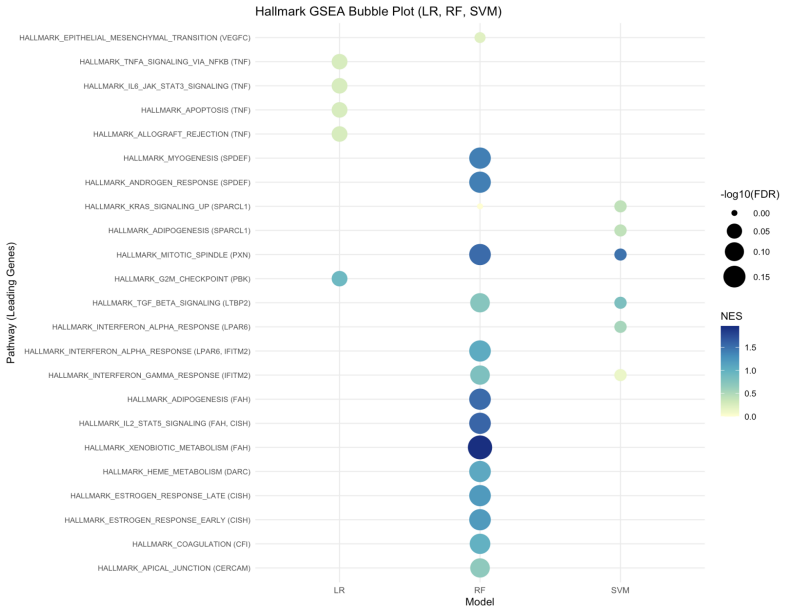


Fig. 26 Enriched pathways found using Hallmark gene set with GSEA.

Although no pathways reached statistical significance in this analysis, we identified several recurring leading genes across different enriched pathways, which we selected for further investigation. Additionally, we highlighted the **Hallmark xenobiotic metabolism** pathway, as it showed the highest enrichment score among those with the highest FDR values (Figure 26).

5.2.3 Important genes

We focused on some important genes, we chose from our previous analyses, to research further. The genes we higlighted are:

- **GALNT10**, as it appeared in the feature lists of all three models.
- **TNF** and **CD1B**, identified through enriched pathways based on the Logistic Regression model's top features.
- **SPARCL1**, **LPAR6** and **IFITM2**, which were highlighted in pathways enriched from the SVM feature list.
- **FAH**, **SPDEF** and **WNT7B**, found in pathways enriched using features selected by the Random Forest model.

6 Discussion

6.1 Interpretation of model performance

The models trained solely on clinical data consistently outperformed all other configurations across all metrics. This outcome is expected, given that cancer stage is

clinically defined using variables such as tumor size (T), lymph node involvement (N), and presence of metastasis (M). These features are directly incorporated into the TNM staging system, which forms the basis of clinical annotations used in our dataset [37]. Therefore, it is not surprising that machine learning models trained on this information were able to achieve near-perfect classification. Clinical features in this case are not only relevant but are the ground truth for the target variable, which naturally gives clinical-only models a strong advantage.

Interestingly, integrating omics data with clinical features led to a slight decrease in model performance. One possible explanation for this is the high dimensionality and complexity of omics data, particularly gene expression and DNA methylation, which can introduce noise and lead to overfitting, particularly when the sample size is modest relative to the number of features [38]. Unless the added data contributes meaningful, non-redundant signal, it can degrade model generalizability. This effect has been reported in similar studies; for example, Poirion et al. observed that adding omics data to clinical variables in survival prediction models did not always improve performance and sometimes even reduced it [5]. In our case, the omics features likely did not add new information that was directly aligned with staging criteria, but rather introduced complexity that diluted the impact of the clinical features.

Models trained using only omics data performed the worst, with metrics close to random classification. This outcome reflects the fundamental disconnect between molecular data and clinical staging definitions. Cancer stage is a measure of disease spread and anatomical progression [37], whereas gene expression and methylation data capture cellular-level activity, transcriptional regulation and epigenetic patterns. While there is undoubtedly a relationship between molecular state and disease severity, it is not necessarily linear or easily captured by classical machine learning models. Furthermore, tumor heterogeneity and inter-patient variability can obscure consistent molecular patterns associated with stage. Similar challenges were noted in the study by Yang et al., where transcriptomic data showed limited prognostic power for colorectal cancer compared to microbiome features [7].

Overall, these findings suggest that while omics data offer rich biological insights, they may not be sufficient on their own for clinical classification tasks such as cancer staging, especially without more refined molecular labels.

6.2 Interpretation of annotated features

To better understand the biological context of the features prioritized by our machine learning models, we performed pathway enrichment analysis on the top genes selected from models trained with omics data. Our goal was to explore whether any of these data-driven features were linked to known cancer-related processes.

The **bronchus development pathway**, while not directly linked to breast cancer, involves the FGF10–FGFR2b signaling axis, which plays a key role in breast development and has been shown to act oncogenically in approximately 10% of breast cancer cases. Interestingly, this pathway also appears in transcriptomic profiles of lung adenocarcinoma patients with a history of breast cancer [39].

The **interferon alpha response pathway** is upregulated in inflammatory breast cancer (IBC), a rare and aggressive subtype. Although its role in IBC remains speculative, it is known to contribute to apoptosis, cellular senescence, increased migration and drug resistance [40]. In estrogen receptor (ER)-positive breast cancer, interferon alpha signaling has been associated with resistance to CDK4/6 inhibitors, which are drugs that block cell cycle progression to limit tumor growth [41].

Leukocyte transendothelial migration (LTM) plays a key role in immune cell infiltration into tumors or metastatic sites. Dysregulation of this process can affect tumor progression and immune evasion. LTM-related genes have been found to be differentially expressed across cancer types and are associated with prognosis and response to immunotherapy [42]. In particular, inhibition of ICAM1 interactions—a central component of this pathway—has been shown to reduce breast cancer cell transendothelial migration and inhibit lung metastasis in experimental models [43].

The **hematopoietic cell lineage pathway** is primarily involved in the formation of blood and immune cells. While not specifically enriched in breast cancer, it may reflect the broader immune context in tumor microenvironments [44].

Similarly, the **amoebiasis pathway** lacks a direct link to breast cancer but includes molecular mechanisms such as amoeboid migration. This cell movement strategy is known to be exploited by invasive breast cancer cells as they metastasize through the extracellular matrix [45].

The **Hallmark xenobiotic metabolism pathway** also emerged with a high enrichment score. Breast cancer cells, particularly in triple-negative subtypes, have been reported to upregulate xenobiotic metabolism genes, resembling the cellular response to toxins in normal cells. This activation may play a role in therapeutic resistance or altered drug processing [46].

Finally, the gene **GALNT10** was identified in the feature lists of all three models. While its precise function in breast cancer remains unclear, the Human Protein Atlas indicates its expression in breast tumor tissues. GALNT10 has been implicated in promoting malignancy in other cancers such as ovarian and liver, and may contribute to immune-suppressive tumor microenvironments [47].

While not all pathways or genes were directly linked to breast cancer staging, several findings align with existing biological knowledge and may reflect processes related to tumor progression, immune evasion, or treatment resistance.

7 Conclusion

In this project, we evaluated the effectiveness of machine learning models for breast cancer stage prediction using gene expression, DNA methylation and clinical data from TCGA. Our results showed that models trained solely on clinical features achieved the highest performance, followed by models combining clinical and omics data, while models trained on omics data alone performed the worst. This trend is likely because cancer staging is defined using clinical and anatomical parameters, which are not directly captured at the molecular level. Pathway enrichment analysis of the top features from omics-based models revealed several biologically relevant processes and genes. While these findings provide insight into potential mechanisms involved in tumor progression

and immune interaction, they are exploratory and limited by the weak performance of the underlying models. Taken together, our work highlights both the potential and current limitations of multi-omics approaches for cancer stage prediction. In our models, integrating gene expression and methylation data with clinical features did not improve performance and omics-only models performed poorly. These results underline that conventional machine learning approaches may not be sufficient to capture subtle molecular patterns related to clinical staging. Limitations such as small sample size, high feature dimensionality, and model simplicity likely contributed to the weak performance of omics-based models. Other studies have reported improved outcomes using larger datasets and more sophisticated methods such as deep learning and multi-omics integration frameworks [5–7], . This suggests that although molecular data alone may be insufficient in basic setups, more advanced strategies could reveal its predictive potential.

Author Contribution

All authors contributed equally to this project.

- Leon was responsible for data preprocessing, differential methylation analysis, and implementing the machine learning models. This included preparing the methylation, expression and clinical data, selecting informative features from differential analyzes, and evaluating the classification performance across different algorithms.
- Payal led the exploratory data analysis and quality control, ensuring the datasets were consistent, clean, and reliable for downstream analysis. She also contributed significantly to the overall structure and writing of the report.
- Renk performed the differential gene expression analysis, performed gene annotation, and carried out pathway enrichment analyses. Additionally, she was responsible for interpreting these results and writing the corresponding sections of the report.

AI Usage

In this project, AI was used to generate code for plotting results, create tables for the report, correct grammar, assist with debugging, understand how to use specific libraries, and compare different libraries in terms of efficiency before choosing which to use.

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